

Document Title: Technology Stack	Date: 15 March 2026
Project Name: ComicCraft - AI Comic Story Creator using Gemini Models	Team ID: SWTID-2026-5650
Maximum Marks: 2 Marks	Status: Approved

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements for ComicCraft - AI Comic Story Creator using Gemini Models.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js / Next.js, Tailwind CSS, Konva.js / HTML5 Canvas	Delivers an interactive, highly responsive web UI for drag-and-drop comic panel layout, real-time speech bubble manipulation, multi-resolution preview, and fast client-side rendering.
2	Backend / Server-Side	Python 3.12 / FastAPI, Celery & Redis Queue	Lightweight, high-performance asynchronous REST API framework ideal for orchestrating Google Gemini AI requests, background batch image processing queues, and microservices.
3	Database / Data Storage	PostgreSQL, Redis, Cloud Object Storage (GCS / S3)	PostgreSQL manages relational project metadata, user profiles & panels; Redis handles rapid session caching & generation status; Cloud Storage holds rendered comic panels & PDFs.
4	Cloud / Hosting / Deployment	Google Cloud Platform (GCP), Docker & Cloud Run	Provides native low-latency integration with Google Gemini infrastructure, scalable containerized execution, auto-scaling web instances, and cost-effective serverless hosting.
5	Version Control & CI/CD	GitHub & GitHub Actions	Ensures distributed team code management, automated linting, unit test execution pipelines, and continuous deployment triggers to Cloud Run on pull requests.
6	Third-Party APIs / Other Tools	Google Gemini API (1.5 Pro/Flash & Imagen 3), ReportLab / PyPDF2, Stripe API	Gemini 1.5 powers story scripting & panel composition; Imagen 3 renders visual panel art; ReportLab compiles high-res PDF comics; Stripe processes creator subscriptions.